Dear Hiring Manager,

I am excited to apply for the AI/Machine Learning Engineer position at Siemens Energy. With a Master's degree in Computer Science specializing in Artificial Intelligence and Machine Learning from Sapienza University of Rome, along with hands-on experience developing machine learning, NLP, and Generative AI applications, I am eager to contribute to Siemens Energy's mission of building innovative digital solutions that support a more sustainable future.

During my master's thesis, I designed and implemented end-to-end machine learning pipelines to evaluate Vision Transformer models under continual learning settings using Python and PyTorch. This experience strengthened my understanding of model training, evaluation, performance benchmarking, and developing reproducible ML workflows. As the author of an IEEE conference publication, I also gained experience analyzing machine learning models, interpreting results, and communicating technical findings effectively.

Beyond research, I enjoy building practical AI applications. I developed DocDelta, an AI-powered document intelligence platform that uses transformer embeddings and semantic search to compare documents and detect meaningful changes. The application exposes AI functionality through FastAPI REST APIs and is containerized with Docker and deployed on AWS, giving me practical experience across the complete machine learning development lifecycle—from model integration to deployment.

My projects have allowed me to work extensively with Python, PyTorch, TensorFlow, Scikit-learn, NumPy, Pandas, FastAPI, Docker, Git, and AWS. I have experience building machine learning pipelines, implementing NLP workflows, developing backend APIs, and writing clean, modular, and maintainable code. I am particularly interested in Generative AI technologies and continue expanding my knowledge of LLMs, retrieval-based systems, and modern AI application development.

What excites me most about Siemens Energy is the opportunity to apply AI to solve real engineering challenges while contributing to technologies that support the global energy transition. The chance to work with experienced engineers on machine learning, NLP, and AI-powered digital products aligns perfectly with my technical interests and long-term career goals.

Thank you for considering my application. I would welcome the opportunity to discuss how my technical background, problem-solving skills, and enthusiasm for applied AI can contribute to Siemens Energy's Digital Products and Solutions team. I look forward to hearing from you.

Sincerely,

Swati Aggrawal